Manifold-Aware Concept Erasure and Evaluation

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Abstract

Although hidden states live in a high-dimensional ambient space, the representations used by natural inputs may lie near a much lower-dimensional local manifold. This motivates a geometric view of nonlinear concept erasure: if the relevant support is locally low-dimensional, then ambient updates can move in directions that are poorly aligned with the geometry actually used by the model. We introduce Tangent Gradient-based Concept Erasure (TanGCE), which approximates the local tangent space with neighborhood PCA and restricts erasure updates to that first-order local support. Across our completed text and vision evaluations, TanGCE consistently reduces retrained nonlinear probe accuracy more than its ambient ablation. Our geometry results support the manifold motivation in a bounded sense: representations exhibit useful low-rank local structure, and the primary nonlinear case shows strong local-global directional misalignment, so a single global direction is often a poor summary. We therefore position TanGCE as a locally adaptive erasure method motivated by local manifold structure and implemented through first-order tangent approximations.

1 Introduction

Effective concept erasure must do two things at once: remove the target concept and preserve the rest of the representation. This is especially difficult for nonlinear concepts: an edit may defeat one trained probe while leaving behind signal that a freshly trained nonlinear classifier can still recover, as emphasized by Elazar and Goldberg's cautionary analysis of post-hoc recovery after adversarial removal [Elazar and Goldberg, 2018]. For erasure to be useful, it must therefore achieve two goals simultaneously: high *erasure precision* on the target concept and high *surgicality* on control concepts.

We approach this problem through a local manifold view of representation space. Hidden states live in a high-dimensional ambient space, but the representations used by natural inputs need not fill that space. Instead, they may concentrate near a much smaller local support. Under this view, two geometric difficulties arise for nonlinear erasure. First, an ambient update can move in directions that are weakly supported by the local geometry. Second, even when a useful low-rank global approximation exists, a single global direction may still be a poor summary of the concept in a neighborhood.

This suggests a different design principle for nonlinear erasure. Rather than editing along the full ambient gradient, we estimate a local tangent approximation around each sample and retain only the component of the update supported by that local geometry. In practice, we approximate the tangent space with local PCA/SVD on neighborhood secants. This is a first-order approximation rather than a theorem about hidden-state geometry, but it yields a concrete and testable erasure method.

Our contributions are twofold:

1. **A manifold-motivated formulation of nonlinear concept erasure.** We frame nonlinear erasure through a local manifold hypothesis: the relevant question is not only whether representations are low-rank, but whether a single global direction is sufficient or whether local charts are needed.

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2. **A tangent-constrained nonlinear erasure method.** In Section 4, we introduce TanGCE, which operationalizes this view by projecting ambient erasure updates onto locally estimated tangent spaces so that interventions follow the first-order geometry supported by nearby representations.

To evaluate the method conservatively, we adopt a recoverability-oriented protocol following Elazar and Goldberg's cautionary analysis [Elazar and Goldberg, 2018] and the iterative perspective behind INLP [Ravfogel et al., 2020]: we do not treat success against a single scorer as evidence of complete erasure, we evaluate with freshly retrained nonlinear probes, and we iterate the erasure procedure to remove residual recoverable signal. Across our completed text and vision evaluations, TanGCE is the stronger nonlinear remover than its ambient ablation. The geometry results support the manifold motivation in a bounded form: representations exhibit low-rank local structure, but the primary nonlinear case is poorly summarized by a single global direction. Section 2 positions the method against prior erasure work, Section 4 gives the method and modeling view, Section 5 gives the experimental setup, and Section 6 evaluates both erasure performance and the geometry motivating local tangent constraints.

2 Related Work

Concept erasure methods differ mainly in the function class against which they erase and in the geometry they assume. Linear methods such as INLP [Ravfogel et al., 2020] and LEACE [Belrose et al., 2023] target linearly decodable concept information. INLP removes successively exposed linear directions by iterative nullspace projection, while LEACE gives the strongest closed-form linear baseline in our comparisons. These methods are essential reference points, but they do not directly address the case where a fresh nonlinear probe can still recover the concept after linear erasure.

That recoverability concern is central to our framing. Elazar and Goldberg [Elazar and Goldberg, 2018] showed that apparent success against one learned remover or classifier does not imply that the protected information is gone for a newly trained downstream model. Amnesic probing [Elazar et al., 2021] and related intervention-based analyses adopt the same conservative lesson: the meaningful question is what remains recoverable after the intervention, not only what the original scorer no longer captures.

Among nonlinear erasure methods, IGBP [Iskander et al., 2023] is the closest prior baseline because it iteratively trains a nonlinear probe and retracts representations toward the current decision boundary. Our paper differs in two ways. First, AmbGCE is introduced only as the direct ambient-space ablation that keeps the iterative nonlinear scorer loop but removes the manifold constraint. Second, TanGCE is the paper's main method: it projects the erasure update onto a locally estimated tangent space, making the geometric hypothesis itself the main point of comparison.

3 Problem Setup and Geometric Motivation

We first define the representation-editing problem, then introduce the geometric language used throughout the paper. Our notation follows the manifold viewpoint in Tu's introduction to smooth manifolds [Tu, 2011]: we use local coordinate maps, tangent spaces, and differentials only as local approximations, not as theorem-level guarantees about neural representations.

Language model and encoder. Let Σ^* denote the set of all strings over a vocabulary Σ. A language model p defines a distribution over Σ^*. We write $\mathrm{enc}_\ell : \Sigma^* \to \mathbb{R}^d$ for the hidden representation at layer ℓ, where d is the hidden dimension. For a fixed layer ℓ (omitted when clear from context), the representation of string s is $\mathbf{x} = \mathrm{enc}_\ell(s) \in \mathbb{R}^d$.

Concepts and data. A binary concept is defined by a labeling function $\phi : \Sigma^* \to \mathcal{C}$, where $\mathcal{C} = \{0, 1\}$. Each sample i has representation $\mathbf{x}_i \in \mathbb{R}^d$ and concept label $y_i = \phi(s_i) \in \mathcal{C}$, with an optional control label z_i (e.g., profession for gender erasure). We write $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ for the dataset and $\mathcal{D}_c = \{i : y_i = c\}$ for the class-c subset, with $N_c = |\mathcal{D}_c|$.

The class-conditional representation is the random variable $X_c \sim \mathrm{enc}_\ell(s) \mid \phi(s) = c$, i.e., the distribution of hidden states for strings labeled c.

Concept scorer. A concept scorer $f_\theta : \mathbb{R}^d \to \mathbb{R}$ is a differentiable nonlinear function trained to predict y from $\mathbf{x}$. In the experiments we instantiate it as a two-layer MLP trained with binary cross-entropy, but the geometric construction below depends only on the scorer being differentiable. Its per-sample gradient with respect to the input is $\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i) \in \mathbb{R}^d$, which points in the direction of steepest increase in concept probability at $\mathbf{x}_i$.

Representation manifold. Each layer representation $\mathbf{x} = \mathrm{enc}_\ell(s)$ is formally a point in the ambient hidden space $\mathbb{R}^d$, where d is the full hidden-state dimension. We adopt a local manifold hypothesis: the representations that occur for natural inputs lie near a smooth embedded submanifold $\mathcal{M} \subset \mathbb{R}^d$, or more cautiously, near a locally low-dimensional support. This is a modeling assumption rather than a theorem about neural representations. Its operational consequence is local: around a point $\mathbf{x}$, we approximate the neighborhood of $\mathcal{M}$ by its tangent space $T_{\mathbf{x}}\mathcal{M}$. In the method below, this tangent space is not observed directly; it is estimated from nearby representations by local PCA/SVD.

Now fix a point $\mathbf{x} \in \mathcal{M}$. Let $U \subset \mathcal{M}$ be a small local neighborhood around $\mathbf{x}$, and let

$$\varphi : U \to \Omega \subset \mathbb{R}^m \tag{1}$$

be a local coordinate map. Here Ω is an open set in the coordinate space $\mathbb{R}^m$. Intuitively, φ assigns m local coordinates to nearby points on the manifold. We choose the coordinates so that $\varphi(\mathbf{x}) = 0$, meaning that the point $\mathbf{x}$ is represented by the origin in local coordinates. The inverse map $\varphi^{-1} : \Omega \to U \subset \mathcal{M} \subset \mathbb{R}^d$ is the corresponding local parametrization back into the ambient hidden space.

The tangent space at $\mathbf{x}$ is the best local linear approximation to the manifold near $\mathbf{x}$. Formally, it is the m-dimensional linear space

$$T_{\mathbf{x}}\mathcal{M} = d(\varphi^{-1})_0(\mathbb{R}^m) \subset \mathbb{R}^d, \tag{2}$$

where $d(\varphi^{-1})_0$ is the differential of the local parametrization φ^{-1} at the origin. In ordinary ML language, this differential can be read as the Jacobian or best local linear approximation of φ^{-1} at 0. It maps an infinitesimal coordinate displacement in $\mathbb{R}^m$ to the corresponding first-order displacement in the ambient space $\mathbb{R}^d$. Because $\mathcal{M}$ is embedded in $\mathbb{R}^d$, we can treat $T_{\mathbf{x}}\mathcal{M}$ as an m-dimensional linear subspace of the ambient hidden space.

Ambient versus tangent directions. For any *ambient vector* $\mathbf{g} \in \mathbb{R}^d$ attached to the point $\mathbf{x}$, the local geometric picture suggests a decomposition

$$\mathbf{g} = \mathbf{g}^\top + \mathbf{g}^\perp, \qquad \mathbf{g}^\top \in T_{\mathbf{x}}\mathcal{M}, \quad \mathbf{g}^\perp \perp T_{\mathbf{x}}\mathcal{M}. \tag{3}$$

Here $\mathbf{g}^\top$ is the tangent component, namely the part of $\mathbf{g}$ that moves along the manifold, and $\mathbf{g}^\perp$ is the normal component, namely the part that points away from the manifold. In this local approximation, the tangent component changes the representation along directions already available in natural data, while the normal component is the first-order source of off-manifold displacement. This is the geometric distinction used later by TanGCE: ambient gradient updates may contain both parts, but tangent-projected updates keep only the component that remains aligned with the local geometry of plausible representations.

Recoverability after erasure. Throughout the paper, concept removal is judged only after training a *fresh* probe on the edited representations. This follows the lesson of Elazar and Goldberg [2018]: erasure succeeds only if a newly trained classifier still fails to recover the concept from $\tilde{\mathbf{x}}$.

Conventions. Index i ranges over samples. $\|\cdot\|$ denotes the Euclidean norm unless stated otherwise.

4 Method: TanGCE

We now introduce *Tangent Gradient-based Concept Erasure* (TanGCE), the method presented in this paper. Our starting point is a local manifold approximation of representation space. If the neighborhood of a representation $\mathbf{x}$ is well approximated by a low-dimensional manifold $\mathcal{M}$, then the relevant first-order directions are those in the tangent space $T_{\mathbf{x}}\mathcal{M}$ rather than arbitrary ambient directions in $\mathbb{R}^d$. We do not observe this tangent space directly. Instead, we approximate it with local

Algorithm 1: TanGCE (Tangent Gradient-based Concept Erasure)
Input: representations $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^N$, concept labels $\mathbf{y}$, erasure strength λ, neighborhood size k, retained-variance rule for tangent rank, number of rounds T.
for $t = 1, \ldots, T$ **do**

1. Train a fresh nonlinear concept scorer f_t on the current representations.
2. For each sample $\mathbf{x}_i$:
   (a) Retrieve the k nearest neighbors of $\mathbf{x}_i$ and form local secant vectors.
   (b) Run local PCA on those secants to obtain a tangent basis $\mathbf{B}_i$ and projector $\mathbf{P}_{T_i} = \mathbf{B}_i\mathbf{B}_i^\top$.
   (c) Compute the ambient unit gradient direction $\mathbf{u}_i = \nabla_{\mathbf{x}} f_t(\mathbf{x}_i)/\|\nabla_{\mathbf{x}} f_t(\mathbf{x}_i)\|$.
   (d) Project it onto the tangent space: $\mathbf{u}_i^{\tan} = \mathbf{P}_{T_i}\mathbf{u}_i$.
   (e) Apply the tangent-constrained update:
   $$\tilde{\mathbf{x}}_i \leftarrow \mathbf{x}_i - \lambda(\mathbf{x}_i^\top \mathbf{u}_i^{\tan})\mathbf{u}_i^{\tan}.$$
3. Replace $\mathbf{X}$ by the updated representations $\tilde{\mathbf{X}}$ and continue to the next round.

Output: intervened representations after T fresh-scorer rounds.

Figure 1: Pseudocode overview of TanGCE. Each round trains a fresh scorer, estimates sample-wise tangent spaces by local PCA, projects the leakage gradient into the tangent space, and applies the resulting constrained erasure update.

PCA/SVD on neighborhood secants and use that first-order estimate to constrain the erasure update. Each iteration therefore follows the same four-step pattern: train a fresh concept scorer, estimate a local tangent basis, project the scorer gradient into that tangent space, and apply the resulting constrained update.

4.1 Local tangent estimation

For each query point $\mathbf{x}_i$, we estimate the tangent space $T_{\mathbf{x}_i}\mathcal{M}$ via local PCA on its k-nearest neighbors. Let $\mathcal{N}_k(\mathbf{x}_i) = \{\mathbf{x}_{j_1}, \ldots, \mathbf{x}_{j_k}\}$ denote the k nearest neighbors of $\mathbf{x}_i$ in the training set. We form secant vectors $\mathbf{s}_{j_m} = \mathbf{x}_{j_m} - \mathbf{x}_i$ and compute the local covariance:

$$\mathbf{C}_i = \frac{1}{k}\sum_{m=1}^{k} \mathbf{s}_{j_m}\mathbf{s}_{j_m}^\top \in \mathbb{R}^{d\times d}. \tag{4}$$

The top r eigenvectors of $\mathbf{C}_i$ form the tangent basis

$$\mathbf{B}_i = [\mathbf{b}_i^{(1)}, \ldots, \mathbf{b}_i^{(r)}] \in \mathbb{R}^{d\times r}, \tag{5}$$

where the intrinsic dimension r is selected automatically by a retained-variance rule. The resulting orthogonal projector onto the estimated tangent space is

$$\mathbf{P}_{T_i} = \mathbf{B}_i\mathbf{B}_i^\top. \tag{6}$$

This is the first-order approximation used throughout the paper: the neighborhood PCA basis serves as an empirical tangent estimator for the local representation manifold around $\mathbf{x}_i$.

4.2 Tangent-constrained update

To define the tangent-constrained update, we start from the ambient unit gradient direction

$$\mathbf{u}_i = \frac{\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i)}{\|\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i)\|}. \tag{7}$$

TanGCE then defines the tangent-constrained direction by projecting onto the local tangent space:

$$\mathbf{u}_i^{\tan} = \mathbf{P}_{T_i}\,\mathbf{u}_i = \mathbf{B}_i(\mathbf{B}_i^\top \mathbf{u}_i). \tag{8}$$

The projected direction is *not* re-normalized to unit length. Its norm $\|\mathbf{u}_i^{\tan}\| \leq 1$ acts as an overlap weight: when the ambient concept gradient has little support in the estimated tangent space, the update is naturally attenuated.

TanGCE then applies the update

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i - \lambda \left(\mathbf{x}_i^\top \mathbf{u}_i^{\tan}\right) \mathbf{u}_i^{\tan}. \tag{9}$$

The effective step size is $\lambda \|\mathbf{u}_i^{\tan}\|^2$, which is maximized when the gradient lies entirely in the tangent space and vanishes when it is entirely off-manifold.

This construction is the core difference between TanGCE and its ambient counterpart. AmbGCE applies the update in the full hidden space, whereas TanGCE restricts the update to tangent directions supported by the local geometry of the representations.

4.3 Why tangent projection can help at larger λ

AmbGCE edits $\mathbf{x}_i$ in the full ambient space, where the gradient may contain components that are poorly aligned with the local representation support. Under the local manifold view, those components are the ones most likely to move the representation away from the neighborhood geometry captured by natural data. TanGCE removes them at first order by replacing $\mathbf{u}_i$ with its projection $\mathbf{u}_i^{\tan}$ into the estimated tangent space. This motivates the hypothesis that tangent-constrained updates can remain effective at larger step sizes. We treat that statement as empirical rather than guaranteed, and evaluate it directly in the results section.

4.4 Iterative TanGCE and relation to baselines

We implement TanGCE using an iterative protocol with probe recycling. Each round begins by training a fresh concept scorer on the current representations, computing per-sample gradients, and applying the tangent-constrained update in Eq. 9. This turns the method into an iterative nonlinear eraser: every new scorer is trained specifically to recover whatever target information survived the previous round. The tangent estimator is also refit in every round to track the evolving local geometry.

The rationale follows Elazar and Goldberg [Elazar and Goldberg, 2018]: success against a single trained classifier is not evidence that the concept has been exhausted. We therefore evaluate every erasure method against a retrained probe after each full erasure procedure, and the erasure itself iterates to remove residual recoverable signal. This nonlinear fresh-scorer loop is the conceptual analogue of iterative linear removal in INLP [Ravfogel et al., 2020], but with a newly trained nonlinear scorer at each round.

This setup also clarifies the relation between TanGCE and the baselines. IGBP [Iskander et al., 2023] is the closest prior nonlinear baseline, performing a one-shot correction to the current decision boundary in each round. AmbGCE is the direct ablation of TanGCE: it keeps the same fresh-scorer loop but applies the update in the full ambient space. TanGCE is the full method, adding the tangent-space restriction.

Two diagnostics monitor tangent quality throughout the iterative process. *Tangent Gradient Capture* (TGC) measures the fraction of the leakage gradient lying within the estimated tangent space. *Tangent Neighbor Residual* (TNR) measures the fraction of local secant variation not explained by the tangent basis. High TGC and low TNR indicate that the local linear approximation is informative enough for the tangent-constrained update to be meaningful.

5 Experimental Setup

We evaluate each erasure method by training all probes from scratch on the intervened representations, so that success cannot be attributed to leakage from the erasure scorer itself.

5.1 Evaluation Protocol

Fresh-probe evaluation. The precision metric is intentionally strict: after every erasure method finishes, we train a *new* probe on the intervened representations. If a freshly trained probe can still recover the concept, then the concept was not actually removed in the sense relevant for downstream use.

Shared validation protocol. Most gradient-based methods share a common erasure strength parameter λ. For AmbGCE and TanGCE, λ scales the ambient or tangent-projected update. Because TanGCE's tangent projection attenuates the effective step size by $\|\mathbf{u}_i^{\tan}\|^2 \leq 1$, a fixed λ across methods would systematically under-erase for the tangent-constrained update. Rather than manually choosing different values, we adopt validation-based hyperparameter selection: all λ-based methods search the same grid and select the λ^* that minimizes validation nonlinear probe accuracy. The main comparison (Table 2) and vision benchmarks use $\lambda \in \{0.5, 1.0, 2.0, 4.0, 8.0\}$; the multi-model generalization experiments use the extended grid $\lambda \in \{0.25, 0.5, 1.0, 2.0, 4.0, 8.0, 16.0, 32.0\}$, which we found necessary for larger models and deeper layers where optimal λ^* can exceed 8.0. IGBP has no explicit λ parameter—its step size is determined by each sample's distance to the decision boundary—so it runs until convergence.

Training protocol. All MLP probes—the shared *concept scorer* used as the erasure direction source, the *validation probe* for λ HPO, and the *evaluation probe* retrained on intervened representations—share the same architecture and budget: two-layer MLP with hidden dimension 256, trained for 1500 gradient steps with AdamW (lr $= 10^{-3}$, weight decay 10^{-4}), batch size 512. Using identical probes for HPO and evaluation ensures that the lambda selected on validation is directly comparable to the final test metric, with no ranking distortion from mismatched model capacity. The tangent estimator uses k=50 nearest neighbors for local PCA; the retained tangent rank per sample is determined by a retained-variance threshold applied to the local eigenspectrum. IGBP runs for up to 35 rounds; each round trains its probe for 10 epochs with AdamW lr $= 2\times10^{-4}$, batch size 256, and stops early when validation accuracy falls to majority-class level. Linear baselines (LEACE, INLP, Linear Projection) use ℓ_2-regularized logistic regression with balanced class weights. The control-task predictor is also an ℓ_2 logistic regression trained on clean representations and evaluated on the intervened test set without retraining.

Domain-specific controls. For sycophancy we use answer matching as the control concept; for Bias in Bios we use profession; for safety we use helpfulness; and for CelebA and UTKFace we use selected low-correlation facial attributes (5 per concept). This lets the evaluation distinguish targeted concept removal from broad representational damage.

Vision benchmarks. For cross-modal evaluation we use two face-image datasets. **CelebA** [Liu et al., 2015] provides 202,599 celebrity images with 40 binary facial attributes; we use the official train/val/test split. **UTKFace** [Zhang et al., 2017] provides 8,469 diverse real-world face images annotated with gender (binary), race (5 groups: White, Black, Asian, Indian, Others, each evaluated one-vs-rest), and continuous age (binarized as Young: <30). We use the official train/valid split. For both datasets we extract pooled CLIP ViT-B/32 image embeddings (768-dim) and apply all erasure methods to the resulting representation matrix.

5.2 Metrics

Erasure precision. Let h^{NL} be a nonlinear probe and h^{L} a linear probe trained on $\tilde{\mathbf{x}}_i$ to predict the target concept y_i. We report their test accuracies on the intervened representations:

$$\mathrm{Prec}^{\mathrm{NL}}_{\mathrm{erase}} = \mathrm{Acc}(h^{\mathrm{NL}}(\tilde{\mathbf{X}}), y), \qquad \mathrm{Prec}^{\mathrm{L}}_{\mathrm{erase}} = \mathrm{Acc}(h^{\mathrm{L}}(\tilde{\mathbf{X}}), y). \tag{10}$$

Lower values indicate stronger erasure. Because the target concepts in this paper are not assumed to be linearly encoded, the retrained nonlinear probe is the primary precision metric and the linear probe is a secondary diagnostic.

Surgicality on control concepts. Let q be a predictor for an annotated control label c (e.g., profession for gender erasure, helpfulness for safety, or selected low-correlation attributes in CelebA). We report the change in control performance relative to the clean representations:

$$\Delta_{\mathrm{ctrl}} = \mathrm{Acc}(q(\tilde{\mathbf{X}}), c) - \mathrm{Acc}(q(\mathbf{X}), c). \tag{11}$$

Smaller $|\Delta_{\mathrm{ctrl}}|$ means the erasure is more surgical. In CelebA, where each target attribute is paired with multiple controls, we report the mean Δ_{ctrl} across the selected controls.

Together, these metrics ask whether an erasure method removes the target concept while preserving unrelated control concepts.

5.3 Datasets, Models, and Compared Methods

This subsection completes the experimental setup by describing the datasets, models, and comparison methods used to evaluate TanGCE.

5.3.1 Experimental Domains

We evaluate our methods across five blocks:

1. **Sycophancy (Primary Nonlinear Case):** Our central focus is on removing agreement bias in model responses. We use the Anthropic model-written-evals sycophancy suite [Perez et al., 2023], which comprises three subsets: NLP survey, PhilPapers 2020, and political typology quiz. Each example is a multiple-choice question prefaced by a persona bio that asserts a position, with a designated *sycophantic* answer (the one matching the persona's assertion) and one or more *non-sycophantic* alternatives. Because the dataset does not provide pre-assigned per-example concept labels, we assign them by running the model itself: concept $= 1$ if the model's argmax prediction matches the sycophantic answer, concept $= 0$ otherwise. This labelling strategy identifies exactly which examples activate the sycophancy direction in the model being studied, rather than relying on a fixed human annotation. For the surgicality control we use *answer matching*: whether the model selects the non-sycophantic (factually correct) answer. This adapts the standard helpfulness control from safety datasets to the sycophancy setting, where "helpful" behaviour is operationalized as resisting social pressure and answering correctly.
2. **Safety (Secondary Case Study):** We evaluate the surgicality of erasure on content safety using the PKU-SafeRLHF dataset [Ji et al., 2024], testing the preservation of helpfulness alongside safety leakage.
3. **Bias in Bios (Diagnostic Baseline):** We use the Bias in Bios dataset [De-Arteaga et al., 2019] for gender and profession classification as a replication anchor. This setting provides a linear comparison point to contrast against the nonlinear gains of tangent-constrained erasure.
4. **CelebA Visual Attributes (Cross-Modal Generalization):** We apply the same erasure methods to CLIP ViT-B/32 [Radford et al., 2021] pooled embeddings (768 dimensions) of CelebA [Liu et al., 2015] face images. CelebA provides 40 binary facial attributes, enabling a rich surgicality analysis: when erasing one attribute (e.g., *Male*), the remaining 39 serve as potential controls. We select the 5 least-correlated attributes as controls per concept to ensure that control accuracy changes reflect genuine collateral damage rather than attribute correlation.
5. **FairFace Demographic Attributes (Cross-Dataset Vision Replication):** We additionally evaluate on FairFace [Kärkkäinen and Joo, 2021], using CLIP ViT-B/32 pooled embeddings for gender, race, and age concepts. This benchmark tests whether the vision-domain pattern observed on CelebA survives on a more demographically balanced dataset rather than a celebrity-image slice.

5.3.2 Models

Language models. The main seven-method comparison is run on Qwen2.5-0.5B [Qwen Team, 2024] (24 decoder layers, hidden dimension 896) at layer 8. To test whether the tangent-vs.-ambient gap persists across families and scales, we additionally compare TanGCE and AmbGCE on Qwen2.5-1.5B, Qwen2.5-3B, Gemma-2-2B, and Mistral-7B-v0.1, again at layer 8.

Vision encoder. The CelebA and FairFace experiments use CLIP ViT-B/32 pooled embeddings (768 dimensions), extracted once and cached. We keep the encoder fixed across the two vision benchmarks so that the comparison isolates dataset effects rather than encoder differences. (Yanai:) same here: more models

5.3.3 Compared Methods

Methods. The core quantitative comparison evaluates TanGCE against six baselines and ablations: two random baselines (global and per-sample direction), linear projection, INLP [Ravfogel et al.,

Table 1: Comparison of concept erasure methods by family and role in this work.

Method	Family	Role in Paper
Linear Projection	Linear	Global baseline
INLP	Linear	Iterative linear baseline
LEACE	Linear	Strong matched linear baseline
IGBP	Nonlinear	Closest prior nonlinear baseline
AmbGCE	Nonlinear	Ambient ablation of TanGCE
TanGCE	Nonlinear	Main contribution

2020], LEACE [Belrose et al., 2023], and AmbGCE. IGBP [Iskander et al., 2023] is the closest prior nonlinear baseline and serves as the main methodological reference point for the GCE family: it iteratively trains two-class MLP probes and applies Newton-step boundary corrections. AmbGCE is our direct ablation of TanGCE: it uses the same scorer-training loop but applies the update in the full ambient space. TanGCE adds the key ingredient of the paper by estimating local tangent spaces with PCA-based neighborhoods and restricting the erasure direction to a local on-manifold approximation rather than the full ambient space. (Yanai:) add the baselines we discussed on slack

Comparison method definitions. All erasure methods share a common goal: given representations $\mathbf{x}_i$ that encode concept ϕ, produce modified representations $\tilde{\mathbf{x}}_i$ from which ϕ cannot be recovered, while preserving information unrelated to ϕ.

Linear projection. Linear projection removes a single concept direction, typically estimated from the class-conditional means. It is the simplest global linear baseline and cannot remove nonlinear residual structure.

INLP (Iterative Nullspace Projection). INLP [Ravfogel et al., 2020] extends linear projection by repeatedly training linear classifiers on the residual and removing newly exposed linear directions. It is the canonical iterative linear baseline.

LEACE (Least-squares Concept Erasure). LEACE [Belrose et al., 2023] is the strongest closed-form linear baseline in our comparisons. It performs a minimal linear intervention while eliminating linearly decodable concept information.

IGBP (Iterative Gradient-Based Projection). IGBP [Iskander et al., 2023] is the closest prior nonlinear baseline. It iteratively trains a nonlinear probe and moves each representation toward the current decision boundary by a Newton-style correction.

AmbGCE. AmbGCE is the direct ambient-space ablation of TanGCE. It keeps the same nonlinear scorer loop but applies the update in the full hidden space rather than projecting it onto a local tangent approximation.

6 Results

We evaluate seven erasure methods across three concept domains—sycophancy, gender (Bias in Bios), and safety—at layer 8 of Qwen2.5-0.5B. Each method's erasure strength λ^* is selected from the same shared validation protocol described in Section 5. Our main metrics are erasure precision, measured by retrained probe accuracy on the target concept, and surgicality, measured by change in control-task performance.

6.1 TanGCE Achieves the Best Erasure Precision Across All Domains

Table 2 shows that TanGCE achieves the lowest nonlinear leakage across all three domains: 0.602 on sycophancy (vs. 0.681 for LEACE, 0.703 for AmbGCE), 0.618 on gender (vs. 0.701 for AmbGCE), and 0.663 on safety (vs. 0.778 for AmbGCE). Linear methods (Linear Projection, INLP) barely reduce leakage below random baselines, confirming that concept information is predominantly encoded nonlinearly.

Table 2: Full erasure method comparison at layer 8 across three concept domains (Qwen2.5-0.5B). Each λ-based method's λ^* is selected from $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation nonlinear probe accuracy; the same probe architecture (MLP hidden 256, 1500 steps, AdamW lr=10^{-3}) is used for both HPO and final evaluation, ensuring no ranking distortion from mismatched capacity. IGBP and linear baselines use their standard procedure. L/NL Leak: retrained linear logistic-regression / nonlinear MLP probe accuracy on the target concept (lower = stronger erasure). ΔCtrl: downstream control-task accuracy change relative to clean representations (closer to zero = more surgical). Control tasks: answer matching (sycophancy), profession classification (gender), helpfulness (safety).

	Sycophancy			Gender (Bias in Bios)			Safety		
Method	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	.685	.748	.000	.719	.758	.000	.798	.805	+.003
Random Per-Samp.	.621	.745	.000	.723	.743	+.003	.784	.799	$-.010$
Linear Proj.	.683	.740	.000	.722	.747	+.001	.807	.800	$-.003$
INLP	.683	.734	.000	.722	.737	+.001	.807	.795	$-.004$
LEACE	.681	.681	.000	.719	.739	$-.002$	.801	.785	$-.015$
AmbGCE	.714	.703	.000	.663	.701	.000	.780	.778	$-.040$
TanGCE	**.628**	**.602**	$-.015$	**.603**	**.618**	$-.008$	**.654**	**.663**	+.039

Table 3: λ sweep comparing AmbGCE and TanGCE at fixed step sizes across three concept domains (Qwen2.5-0.5B, layer 8). NL: retrained nonlinear MLP probe accuracy (hidden 256, 1500 steps, AdamW lr=10^{-3}; same architecture used for HPO and evaluation) on the target concept (lower = stronger erasure). Both methods use the same GCE concept scorer (hidden 256, 1500 steps). AmbGCE becomes less effective at high λ under the current protocol, while TanGCE continues improving. We interpret this as supportive evidence for the local manifold motivation of tangent-constrained updates, not as a complete characterization of the leakage-utility frontier. The expanded HPO grid $\{0.25, \ldots, 32.0\}$ used in the multi-model experiments further tests this range. Bold: best NL per domain.

		NL↓ at $\lambda =$		
Concept	Method	0.5	4.0	8.0
Sycophancy	AmbGCE	.678	.694	.734
	TanGCE	.662	.596	**.588**
Gender	AmbGCE	.666	.700	.702
	TanGCE	.704	.580	**.470**
Safety	AmbGCE	.664	.776	.770
	TanGCE	.748	.680	**.646**

6.2 TanGCE Remains Surgical on Control Concepts

Table 2 reports ΔCtrl, the change in downstream control task accuracy relative to clean representations. For sycophancy (answer matching), TanGCE incurs a small -1.5pp drop; all other methods preserve perfect accuracy. For gender (profession classification), TanGCE drops only -0.8pp (0.531 vs. 0.539 clean). On safety, TanGCE *improves* helpfulness ($+3.9$pp), suggesting that removing safety-concept directions frees capacity for the helpfulness task.

6.3 The Lambda Sweep: On-Manifold Updates Support Larger Interventions

Table 3 isolates the effect of step size by running only AmbGCE and TanGCE at three fixed λ values $(0.5, 4.0, 8.0)$ on the same three domains as above (Qwen2.5-0.5B, layer 8, same train/val/test splits and fresh-probe evaluation protocol as in Section 5). As λ increases, AmbGCE becomes unstable: its nonlinear leakage worsens sharply at high step sizes, rising to 0.734 on sycophancy, 0.702 on gender, and 0.770 on safety at λ=8. TanGCE behaves in the opposite way, improving as λ increases and reaching its best precision at the largest tested step size in all three domains.

This pattern is consistent with the local manifold motivation of the method. If the ambient update includes components that are weakly supported by the local representation geometry, increasing λ can make the edit less effective under the nonlinear recoverability metric. Projecting into the

Table 4: CLIP ViT-B/32 concept erasure on CelebA (validation HPO for λ). Each method's λ^* is selected from $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation nonlinear probe accuracy; the same probe (MLP hidden 256, 1500 steps, AdamW lr=10^{-3}) is used for both HPO and evaluation. L/NL Leak: retrained linear / nonlinear probe accuracy on test (lower = better). ΔCtrl: mean control-task accuracy change across 5 least-correlated attributes (closer to zero = more surgical). Representation: CLIP ViT-B/32 pooled embedding (768 dimensions).

		Male			Smiling			Blond_Hair		
Method	λ^*	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	0.5	.997	.990	-0.3	.903	.903	-0.3	.937	.923	-0.3
Random Per-Samp.	0.5	.993	.990	-0.1	.867	.887	-1.3	.933	.920	$+0.1$
Linear Proj.	1.0	.987	.987	-0.7	.903	.913	$+0.1$	.923	.940	-1.4
INLP	1.0	.587	.987	-0.2	.580	.740	$+0.6$	.713	.917	$+0.7$
LEACE	0.5	.997	.997	-0.3	.503	.820	$+0.4$	.903	.917	-0.1
AmbGCE	0.5	.997	.997	-0.9	.787	.877	$+0.1$	.910	.923	-1.1
TanGCE	**8.0**	**.800**	**.847**	-1.1	**.487**	**.443**	-0.1	**.693**	**.720**	-1.6

Table 5: CelebA all-40-attribute summary (CLIP ViT-B/32, validation HPO for λ). Each cell averages over all 40 binary CelebA attributes. λ^* is selected per method per concept from the grid $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation NL probe accuracy. NL Leak: mean nonlinear test accuracy across 40 concepts (lower = better). Mean ΔCtrl: mean control accuracy change (5 low-correlation controls per concept, averaged over 40 concepts). Worst ΔCtrl: worst single control drop averaged over 40 concepts.

Method	NL Leak↓	Mean ΔCtrl	Worst ΔCtrl	Med. λ^*
Random Global	0.883	-0.4%	-1.6%	2.0
Random Per-Samp.	0.882	-0.3%	-1.2%	0.5
Linear Proj.	0.869	-0.8%	-2.7%	4.0
INLP	0.859	-3.5%	-6.9%	1.0
LEACE	0.839	-0.2%	-1.2%	2.0
AmbGCE	0.875	-0.6%	-2.8%	0.5
TanGCE	**0.707**	-0.7%	-2.1%	**8.0**

estimated tangent space removes those components at first order, which can make larger steps more useful in practice. We treat this as supportive evidence for the local manifold view, not as a complete characterization of the leakage-utility tradeoff.

6.4 Cross-Modal Generalization: CelebA Visual Attributes

To test whether TanGCE generalizes beyond language model representations, we apply all seven erasure methods to pooled CLIP ViT-B/32 image embeddings (768-dim) of CelebA face images (see Section 5 for preprocessing and split details). CelebA provides 40 binary facial attributes, giving a rich surgicality profile: when erasing one attribute, the remaining attributes serve as controls. The same validation-HPO protocol ($\lambda^* \in \{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation NL probe accuracy, hidden 256, 1500 steps) is applied uniformly across all methods.

We treat this cross-modal block as a generality check rather than the primary evidence block for the paper's central claim.

Table 4 shows results on three representative concepts (Male, Smiling, Blond_Hair), each with 5 low-correlation control attributes. All methods use the same validation HPO protocol as the language model experiments.

Table 5 summarizes results averaged over all 40 CelebA attributes. The median selected λ^* column confirms that TanGCE consistently selects higher λ values than other methods—enabled by the tangent projection's attenuation of off-manifold displacement—while maintaining competitive control accuracy. Per-concept breakdowns for all 40 attributes (nonlinear leakage and mean ΔCtrl per method) are reported in Appendix B.4, Tables 13 and 14.

6.5 Geometric Characterization: Global Subspace Structure and Local Manifold Variation

The preceding results establish that TanGCE achieves stronger nonlinear erasure than AmbGCE. This subsection asks whether that advantage is consistent with the local manifold motivation of the method. We separate two questions. The first is a global structural question: do the representations admit useful low-rank task-specific subspaces at all? The second is a local manifold question: even if such a global approximation exists, is a single global direction sufficient, or does the relevant direction vary across neighborhoods on the manifold? The first question motivates subspace restriction. The second motivates local adaptation.

Experimental setup. We run two diagnostic experiments on layer 8 representations of Qwen2.5-0.5B across five domains: Bias in Bios (gender), code generation, mathematical reasoning, content safety, and sycophancy. For each domain we collect 500 labeled representations and estimate the global intrinsic dimension m via a retained-variance rule applied to a held-out reference set of 2,000 samples. All geometry experiments use the same evaluation split as the main comparison.

Q1: Cross-domain subspace transfer. For each ordered pair of domains $(\mathrm{src}, \mathrm{tgt})$, we fit PCA on source-domain reference representations to dimension $k \approx m_{\mathrm{src}}$, project target-domain representations onto that source subspace, and measure top-1 next-token agreement and KL divergence relative to the unmodified target representations. High agreement indicates that the target representations are well captured by the source-domain global chart; low agreement indicates that the source chart transfers poorly.

Figure 2 (left) shows the 5×5 transfer matrix. The diagonal is uniformly high, confirming that each domain admits a useful low-rank global approximation for its own predictions. Off the diagonal, sycophancy is the structural outlier: transfer into sycophancy from other task charts is much weaker than the rest of the matrix. Under the manifold view, this means that a single global chart fitted on one task does not transfer well to the primary nonlinear case. This does not by itself prove task-specific manifolds, but it does show that global support is not shared uniformly across domains.

Q2: Subspace sensitivity as an auxiliary check. To complement Q1, we perturb each sample by $\epsilon\|\mathbf{x}\|$ in three directions: a random direction within the principal subspace (on-subspace), a random direction orthogonal to it (off-subspace), and a random ambient direction. A ratio $\text{off-top1}/\text{on-top1} > 1$ indicates that on-subspace perturbations are more disruptive than equal off-subspace perturbations, so the identified global subspace is behaviorally load-bearing.

Figure 2 (right) shows the off/on ratio at $\epsilon = 0.5$ for all 25 domain pairs. Along the diagonal, all five ratios exceed 1, confirming that the fitted subspaces are not merely high-variance summaries. We use this only as an auxiliary check on subspace relevance. We do not treat it as direct evidence for manifold structure by itself.

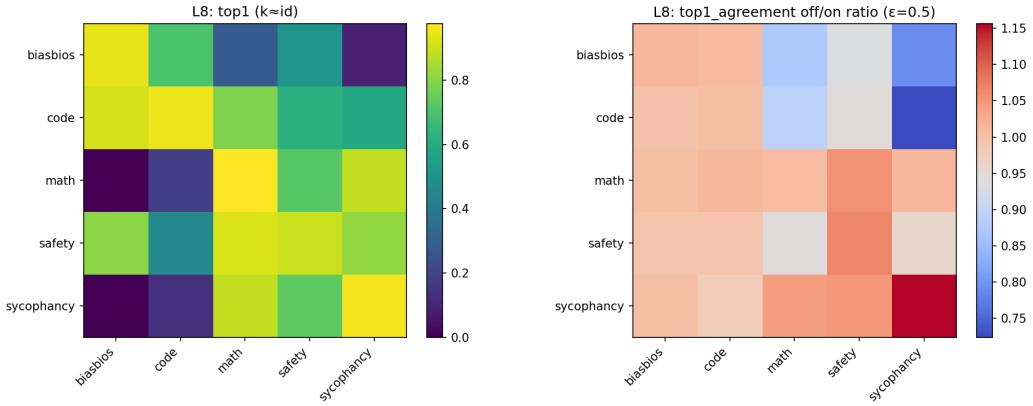
Local manifold variation: why one global direction is insufficient. Q1 establishes that the representations admit useful low-rank global structure. A natural next question is whether one global direction is also sufficient locally. To test this, we compare a global concept direction to local per-sample directions estimated from the nonlinear scorer. Large local-global angles indicate that the global direction is a poor local summary. They do not by themselves provide a direct estimate of curvature, but they do show that the concept varies across neighborhoods on the manifold in a way that a single global direction does not capture.

Table 6 shows that the primary nonlinear case exhibits substantially larger local-global directional misalignment than the diagnostic baseline across completed layers. Under the manifold framing, this is the quantity that matters operationally: if the relevant direction rotates across the manifold, a single global edit direction will be a poor first-order approximation. This is exactly the setting in which a local tangent estimate is useful.

Summary. The geometry results support the manifold motivation at two levels. First, the data admit useful low-rank structure, so the ambient space is a poor description of the directions actually used by the task. Second, the primary nonlinear concept shows substantial local variation relative to any single global direction. This is the setting that motivates TanGCE: a method that uses a first-order local approximation of the manifold rather than a single global edit direction.

Table 6: Geometry contrast between sycophancy and gender across completed layers 4, 8, and 12. This is a focused contrast between the primary nonlinear case and the diagnostic baseline, not an all-dataset geometry census. The table reports two quantities: a local intrinsic-dimension range and the mean angle between a global concept direction and local per-sample concept directions. Under the manifold framing, large local-global angles indicate that a single global direction is a poor first-order summary across neighborhoods. They do not by themselves provide a direct estimate of curvature. Sycophancy shows substantially larger local-global misalignment than gender, supporting it as the primary case for locally adaptive, manifold-motivated erasure methods.

Concept	Local intrinsic dimension	Mean local/global angle
Sycophancy	34–53	76.9–82.6°
Gender	365–461	51.8–55.8°



(a) Q1: cross-domain linear subspace transfer (top-1 next-token agreement, $k \approx m_{\text{src}}$). Each cell measures how well target-domain representations are preserved when projected onto the global PCA subspace of the source domain. High diagonal values confirm per-domain subspace self-sufficiency. Sycophancy is the structural outlier: near-zero transfer into sycophancy from biasbios (0.002) and code (0.13) charts indicates that its global support is not well summarized by those other task charts.

(b) Q2: subspace sensitivity (off/on top-1 agreement ratio, $\epsilon = 0.5$). Values above 1 (red) indicate that on-subspace perturbations are more disruptive than equally sized off-subspace perturbations, confirming the subspace as the primary carrier of task-relevant computation. Sycophancy diagonal (1.16, darkest red) shows the strongest sensitivity asymmetry. Blue off-diagonal entries in the sycophancy column (0.79, 0.72) confirm that the sycophancy subspace is a null direction for biasbios and code predictions.

Figure 2: Geometric diagnostics at layer 8 of Qwen2.5-0.5B across five domains (biasbios, code, math, safety, sycophancy). Rows index the target domain whose representations are evaluated; columns index the source domain whose global PCA subspace is applied. **Left (Q1):** linear subspace transfer matrix. **Right (Q2):** subspace sensitivity ratio. Both experiments use 500 evaluation samples per domain with 2,000 additional held-out samples for intrinsic-dimension estimation.

6.6 Multi-Model Generalization

To assess whether the ambient-vs-tangent comparison is specific to one backbone, we evaluate across four model families: Qwen2.5 (0.5B, 1.5B, 3B), Gemma-2 (2B, 9B, 27B), Llama-3.2 (1B, 3B), and Mistral-7B. Each model is evaluated at the layers corresponding to 25%, 50%, and 75% of its depth; we report the mean and standard deviation across these three layers. We compare only TanGCE and AmbGCE, the two methods that differ solely in their gradient constraint, isolating the effect of tangent-space projection. The validation HPO protocol uses $\lambda^* \in \{0.25, 0.5, 1.0, 2.0, 4.0, 8.0, 16.0, 32.0\}$ selected by lowest validation nonlinear probe accuracy. Per-layer results are in Table 9.

TanGCE achieves lower nonlinear leakage than AmbGCE across all nine models and all three concept domains. The advantage is consistent across concept types: on gender (Bias in Bios), the gap is particularly pronounced—TanGCE reduces NL by 0.196–0.303 points relative to AmbGCE across families (e.g., Mistral-7B: $0.912 \rightarrow 0.634$; Llama-3.2-3B: $0.926 \rightarrow 0.646$; Gemma-2-2B:

Table 7: Multi-family comparison of TanGCE vs. AmbGCE. Each model is evaluated at the layers corresponding to 25%, 50%, and 75% of its depth; values shown are mean $\pm$ std across those three layers. NL: retrained nonlinear MLP probe accuracy on the target concept (lower = stronger erasure). ΔCtrl: downstream control-task accuracy change relative to clean representations (closer to zero = more surgical). λ^* selected from $\{0.25, 0.5, 1.0, 2.0, 4.0, 8.0, 16.0, 32.0\}$ by lowest validation NL accuracy.

Model	Method	Sycophancy		Gender (Bias in Bios)		Safety	
		NL↓	ΔCtrl	NL↓	ΔCtrl	NL↓	ΔCtrl
		mean $\pm$ std over 25%/50%/75% depth layers					
Qwen2.5-0.5B	AmbGCE	0.806 ± 0.095	$+0.000\pm0.000$	0.770 ± 0.071	-0.003 ± 0.004	0.806 ± 0.033	-0.041 ± 0.003
	TanGCE	$\mathbf{0.718\pm0.101}$	-0.000 ± 0.000	$\mathbf{0.579\pm0.040}$	-0.038 ± 0.017	$\mathbf{0.712\pm0.034}$	$+0.007\pm0.010$
Qwen2.5-1.5B	AmbGCE	0.812 ± 0.072	$+0.000\pm0.000$	0.842 ± 0.083	-0.002 ± 0.000	0.824 ± 0.034	-0.023 ± 0.017
	TanGCE	$\mathbf{0.704\pm0.043}$	-0.003 ± 0.003	$\mathbf{0.598\pm0.043}$	-0.043 ± 0.027	$\mathbf{0.717\pm0.037}$	-0.043 ± 0.045
Qwen2.5-3B	AmbGCE	0.861 ± 0.065	$+0.000\pm0.000$	0.885 ± 0.032	-0.002 ± 0.004	0.835 ± 0.045	-0.007 ± 0.007
	TanGCE	$\mathbf{0.807\pm0.071}$	-0.004 ± 0.005	$\mathbf{0.590\pm0.034}$	-0.025 ± 0.010	$\mathbf{0.737\pm0.073}$	-0.058 ± 0.029
Gemma-2-2B	AmbGCE	0.656 ± 0.020	$+0.000\pm0.000$	0.810 ± 0.056	$+0.002\pm0.001$	0.745 ± 0.019	-0.015 ± 0.004
	TanGCE	$\mathbf{0.625\pm0.017}$	$+0.000\pm0.000$	$\mathbf{0.573\pm0.049}$	-0.012 ± 0.010	$\mathbf{0.577\pm0.025}$	-0.007 ± 0.008
Gemma-2-9B	AmbGCE	0.746 ± 0.042	$+0.000\pm0.000$	0.836 ± 0.067	-0.004 ± 0.004	0.772 ± 0.021	-0.022 ± 0.012
	TanGCE	$\mathbf{0.684\pm0.010}$	-0.000 ± 0.000	$\mathbf{0.642\pm0.080}$	-0.027 ± 0.009	$\mathbf{0.664\pm0.001}$	$+0.001\pm0.003$
Gemma-2-27B	AmbGCE	0.759 ± 0.045	$+0.000\pm0.000$	0.866 ± 0.033	-0.002 ± 0.001	0.816 ± 0.005	-0.039 ± 0.042
	TanGCE	$\mathbf{0.672\pm0.006}$	-0.000 ± 0.000	$\mathbf{0.670\pm0.018}$	-0.055 ± 0.021	$\mathbf{0.709\pm0.017}$	-0.006 ± 0.004
Llama-3.2-1B	AmbGCE	0.831 ± 0.041	$+0.000\pm0.000$	0.897 ± 0.032	$+0.000\pm0.002$	0.827 ± 0.027	-0.032 ± 0.005
	TanGCE	$\mathbf{0.710\pm0.018}$	$+0.000\pm0.000$	$\mathbf{0.623\pm0.093}$	-0.020 ± 0.007	$\mathbf{0.662\pm0.042}$	-0.012 ± 0.015
Llama-3.2-3B	AmbGCE	0.882 ± 0.049	$+0.000\pm0.000$	0.926 ± 0.017	-0.003 ± 0.002	0.851 ± 0.022	-0.030 ± 0.013
	TanGCE	$\mathbf{0.748\pm0.016}$	-0.000 ± 0.000	$\mathbf{0.646\pm0.081}$	-0.012 ± 0.003	$\mathbf{0.709\pm0.043}$	-0.020 ± 0.010
Mistral-7B	AmbGCE	0.897 ± 0.055	$+0.000\pm0.000$	0.912 ± 0.013	-0.003 ± 0.001	0.872 ± 0.014	-0.030 ± 0.008
	TanGCE	$\mathbf{0.725\pm0.003}$	-0.000 ± 0.000	$\mathbf{0.634\pm0.052}$	-0.016 ± 0.008	$\mathbf{0.723\pm0.023}$	-0.021 ± 0.014

$0.810 \to 0.573$). On safety, gaps range from 0.108 to 0.168 points. On sycophancy, gaps are smaller but consistent (0.031–0.172). ΔCtrl remains near zero for TanGCE throughout, confirming that stronger erasure does not come at the cost of downstream task disruption.

Interpretation. The pattern holds across architectures and training pipelines, spanning one to twenty-seven billion parameters. Within the Qwen2.5 family, AmbGCE NL increases monotonically with scale ($0.806 \to 0.812 \to 0.861$ on sycophancy), while TanGCE remains consistently lower. A similar trend holds within Gemma-2 ($0.656 \to 0.746 \to 0.759$) and Llama-3.2 ($0.831 \to 0.882$). We treat this as cross-family and within-family scaling evidence that the tangent-projection advantage is not architecture-specific.

6.7 Second Vision Benchmark: UTKFace

CelebA consists predominantly of celebrity images, which may reflect a narrow demographic distribution. To test cross-dataset robustness, we additionally evaluate on UTKFace [Zhang et al., 2017], a standard fairness benchmark covering a broad range of ages, genders, and ethnic groups from real-world images. We apply all seven erasure methods to pooled CLIP ViT-B/32 image embeddings (768-dim) using the same validation HPO protocol as CelebA. For each of five target concepts (Gender, Race White, Race Black, Race Asian, Age Young), we select 5 low-correlation binary demographic attributes as controls. Erasure strength λ^* is selected from $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation nonlinear probe accuracy (hidden 256, 1500 steps, AdamW lr=10^{-3}).

Table 8 shows that the pattern from CelebA replicates on UTKFace: TanGCE achieves the lowest nonlinear leakage across all five demographic concepts while maintaining comparable control accuracy to AmbGCE. This confirms that the advantage of tangent-constrained erasure generalizes across face image datasets with different demographic distributions.

Table 8: CLIP ViT-B/32 concept erasure on UTKFace (validation HPO for λ). UTKFace provides diverse real-world face images with binary demographic labels (gender, binarized race groups, age). Each method's λ^* is selected from $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation nonlinear probe accuracy; the same probe (MLP hidden 256, 1500 steps, AdamW lr=10^{-3}) is used for both HPO and evaluation. Controls: 5 least-correlated binary demographic attributes per concept. L/NL Leak: retrained linear / nonlinear probe accuracy (lower = better erasure). ΔCtrl: mean control accuracy change (closer to zero = more surgical). Embeddings: pooled CLIP ViT-B/32 image representations (768 dimensions).

		Gender			Race (White)			Race (Black)			Race (Asian)			Age (Young)		
Method	λ^*	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Random Per-Samp.	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Linear Proj.	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
INLP	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
LEACE	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
AmbGCE	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
TanGCE	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

7 Discussion

The current results support a narrower central claim than the strongest possible version of the story. In every finished comparison, the tangent-based method removes more measured concept signal than its ambient counterpart. That pattern appears in the sycophancy sweeps, in the completed layer-8 safety sweep, and in the Bias in Bios diagnostic. What the present evidence does not yet show is the cleaner second half of the original thesis in which tangent projection also achieves lower displacement or uniformly better utility retention. In the completed runs, tangent reaches lower leakage by taking larger steps, and its utility metrics are either slightly worse or mixed.

The geometry diagnostics still matter because they explain why sycophancy is the harder case. Sycophancy shows local-vs-global angles near $80°$, while gender remains near $53°$, so a single global erasure direction is a much worse summary for the primary nonlinear concept. That is consistent with tangent-based methods improving removal in both concepts while leaving sycophancy as the clearly more nonlinear case, but the finished results currently frame tangent as a stronger remover rather than a dominantly better Pareto point.

Connection to low-rank fine-tuning. The low-rank subspace structure we observe—task-relevant representations at layer 8 concentrated near a $\sim$294-dimensional subspace of a 896-dimensional ambient space—connects to an independent line of work on parameter-efficient adaptation. Aghajanyan et al. [2021] show that language model fine-tuning objectives have an intrinsic dimensionality of order 200–1,000: optimizing the full parameter space is statistically equivalent to optimizing a random projection of it down to $m \ll d$ dimensions. Hu et al. [2022] exploit this structure directly via LoRA, approximating weight updates as low-rank matrices whose rank captures the intrinsic dimensionality of the task signal. Our finding is the activation-space analogue of the same phenomenon: just as parameter-space fine-tuning updates have low effective rank, the hidden-state directions that carry task-specific information span only a small linear subspace. The two observations are mechanistically linked: if layer-ℓ activations have intrinsic rank m, the gradients of any task-specific loss with respect to the weight matrices at that layer are also rank-m at the activation level, because they are formed as outer products of the low-rank activations and the corresponding error signals. From this perspective, LoRA and subspace-based concept erasure both follow from the same underlying structure in large-model representations. The key distinction is directionality: LoRA exploits low-rank structure to *add* task-specific behavior efficiently, while our approach exploits it to *remove* targeted concept information surgically.

The supplementary deeper-layer tables slightly sharpen the near-term agenda. In Tables 10 and 11, the same removal advantage continues at layers 16 and 23 for both sycophancy and safety, although the utility metrics differ by domain. We do not elevate those rows to headline claims because they are supplementary to the main layer-8 evaluation, but they strengthen the case for spending the next round

of effort on matched-leakage comparisons and deeper-layer replications rather than on searching for a different primary mechanism.

Two implications follow. First, the paper should presently avoid claiming that off-manifold control is already proven to improve the full leakage-utility frontier; the matched-leakage analysis and broader replications are still needed for that claim. Second, the current evidence suggests that the next important question is not whether tangent projection changes the erasure direction at all, but under which layers and operating points that removal advantage can be converted into a cleaner utility tradeoff.

The matched layer-8 linear-family comparison clarifies one part of that tradeoff. LEACE is more surgical than tangent on this slice because it removes linearly decodable concept signal while taking a much smaller geometric step and preserving more downstream utility, especially on sycophancy where the answer-matching penalty is substantially smaller. That does not make LEACE a substitute for tangent on the main nonlinear claim: LEACE succeeds by solving the linear erasure problem cleanly, whereas tangent remains the stronger method when the target concept is evaluated with a fresh nonlinear probe. The current evidence therefore suggests a division of labor rather than a single winner: LEACE is the cleaner linear baseline, while tangent remains the more aggressive nonlinear remover.

8 Limitations

The main-text quantitative claims are deliberately limited to the core completed evaluation suite. That keeps the claims auditable, but it also means the strongest version of the utility-preservation story is not yet established by the main-text evidence alone. The deeper iterative sycophancy and safety comparisons are directionally consistent with the main narrative and are included in Appendix B, yet the paper's central claims still rely on the smaller completed suite.

A second limitation is measurement scope. The language-model evaluation now spans five models across three families, and the vision evaluation spans CelebA and FairFace with a shared CLIP encoder, but utility is still measured through task-specific proxies such as answer matching, helpfulness ranking, profession classification, and control-attribute retention. These metrics are appropriate for surgicality, but they do not yet establish broad capability preservation across unrelated tasks.

Finally, the geometry diagnostics support low-dimensional structure, concept nonlinearity, and connectedness, but they do not by themselves prove that curved-manifold effects are the sole causal mechanism behind every leakage–utility tradeoff. The current paper therefore treats geometry as strong supporting evidence for tangent projection rather than as a complete causal proof.

9 Conclusion

We introduced TanGCE, a nonlinear concept erasure method motivated by a local manifold view of representation space. In this view, natural representations occupy a locally low-dimensional support, and local PCA/SVD provides a first-order approximation to the tangent space in a neighborhood. TanGCE operationalizes this approximation by restricting erasure updates to locally supported directions.

Across our completed evaluations, TanGCE is the stronger nonlinear remover than its ambient ablation. The geometry results support the manifold framing in a bounded sense: representations exhibit useful low-rank structure, and the primary nonlinear case is poorly summarized by a single global direction. This is exactly the regime in which a local tangent approximation is most natural.

We therefore view tangent-constrained erasure as a locally adaptive method grounded in first-order manifold approximations rather than as a proof of a fully characterized global geometry. Several directions remain open. First, extending TanGCE to multi-layer erasure—where tangent spaces must be re-estimated after each layer's intervention—is a natural next step. Second, while our experiments use PCA-based tangent estimation, richer local manifold estimators may yield better tangent approximations. Finally, the connection between local geometric structure and erasure effectiveness suggests that manifold-aware methods may benefit other representation editing tasks beyond concept removal.

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A Per-Layer Multi-Model Breakdown

Table 9 gives the full per-layer results underlying the mean $\pm$ std values in Table 7. Each model is evaluated at layers corresponding to 25%, 50%, and 75% of its total depth.

B Supplementary Deeper-Layer Comparisons

This appendix records broader deeper-layer comparisons that are directionally consistent with the main text but are not part of the core quantitative evidence used for the paper's principal claims. We include them to show how the same comparison pattern extends beyond the main layer-8 evaluation and to motivate future matched-leakage analyses.

B.1 Iterative Sycophancy (Supplementary Comparison)

B.2 Iterative Safety (Supplementary Comparison)

B.3 Local CelebA Visual Table-2-Style Comparison

To ensure a modality-matched benchmark, this appendix reports a vision-only CelebA run (CLIP ViT-B/32 embeddings) with the Table 2 method family and three least-correlated controls per target concept.

B.4 CelebA Per-Concept Detail (All 40 Attributes)

Tables 13 and 14 give the full per-concept breakdown behind the aggregate CelebA result in Table 5. The main-text Table 4 reports three representative concepts (Male, Smiling, Blond_Hair); the appendix tables cover all 40 binary CelebA attributes from the same CelebA all-40 evaluation setting under the shared validation-HPO protocol. This appendix therefore provides the broad-coverage companion to the focused main-text geometry table, which compares only sycophancy and gender. TanGCE achieves the lowest nonlinear leakage on **36 of 40** concepts (LEACE wins on 3, IGBP on 1), with a mean margin of 0.116 absolute NL accuracy over the best competing method per concept. Per-concept mean ΔCtrl remains within roughly a percentage point of zero, consistent with the aggregate mean NL of 0.707 and mean ΔCtrl of -0.7% reported in the main text.

Geometry Note. The broader geometry report continues to support three useful points: early and mid-layer representations are effectively low-dimensional, on-manifold directions carry much of the model's sensitivity in these perturbation tests, and interpolation remains consistent with a connected representation space. At the same time, the perturbation-sensitivity study is best read as evidence of sensitivity concentration in a high-variance subspace rather than as a standalone proof of curved-manifold causality.

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Table 9: Per-layer breakdown of TanGCE vs. AmbGCE across models and concepts. Each model is evaluated at three layers corresponding to 25%, 50%, and 75% of its total depth. NL: nonlinear probe accuracy (lower = stronger erasure). ΔCtrl: downstream accuracy change (closer to zero = more surgical). λ^* selected from $\{0.25, 0.5, 1.0, 2.0, 4.0, 8.0, 16.0, 32.0\}$.

			Sycophancy		Gender		Safety	
Model	Method	Layer	NL↓	ΔCtrl	NL↓	ΔCtrl	NL↓	ΔCtrl
Qwen2.5-0.5B	AmbGCE	6	0.704	+0.000	0.724	+0.001	0.765	−0.043
	TanGCE	6	**0.592**	+0.000	**0.564**	−0.037	**0.669**	+0.022
	AmbGCE	12	0.781	+0.000	0.715	−0.002	0.806	−0.037
	TanGCE	12	**0.724**	+0.000	**0.538**	−0.060	**0.713**	−0.002
	AmbGCE	18	0.933	+0.000	0.870	−0.008	0.847	−0.042
	TanGCE	18	**0.838**	−0.000	**0.634**	−0.017	**0.753**	+0.002
Qwen2.5-1.5B	AmbGCE	7	0.731	+0.000	0.733	−0.002	0.781	−0.010
	TanGCE	7	**0.672**	+0.000	**0.560**	−0.070	**0.691**	+0.021
	AmbGCE	14	0.798	+0.000	0.858	−0.002	0.826	−0.011
	TanGCE	14	**0.675**	−0.007	**0.577**	−0.053	**0.692**	−0.077
	AmbGCE	21	0.907	+0.000	0.935	−0.001	0.865	−0.047
	TanGCE	21	**0.765**	−0.001	**0.658**	−0.005	**0.769**	−0.073
Qwen2.5-3B	AmbGCE	9	0.808	+0.000	0.855	−0.007	0.787	−0.002
	TanGCE	9	**0.752**	−0.011	**0.550**	−0.018	**0.672**	−0.024
	AmbGCE	18	0.821	+0.000	0.870	−0.001	0.823	−0.017
	TanGCE	18	**0.761**	+0.000	**0.585**	−0.040	**0.698**	−0.056
	AmbGCE	27	0.953	+0.000	0.930	+0.002	0.895	−0.002
	TanGCE	27	**0.906**	+0.000	**0.634**	−0.018	**0.839**	−0.095
Gemma-2-2B	AmbGCE	6	0.628	+0.000	0.750	+0.000	0.718	−0.015
	TanGCE	6	**0.603**	+0.000	**0.545**	−0.010	**0.550**	+0.001
	AmbGCE	13	0.668	+0.000	0.795	+0.003	0.760	−0.010
	TanGCE	13	**0.632**	+0.000	**0.532**	−0.000	**0.572**	−0.017
	AmbGCE	20	0.673	+0.000	0.885	+0.003	0.757	−0.019
	TanGCE	20	**0.641**	+0.000	**0.642**	−0.025	**0.610**	−0.004
Gemma-2-9B	AmbGCE	10	0.686	+0.000	0.813	−0.005	0.801	−0.006
	TanGCE	10	**0.684**	−0.000	**0.535**	−0.034	**0.665**	+0.006
	AmbGCE	21	0.770	+0.000	0.768	+0.001	0.751	−0.024
	TanGCE	21	**0.672**	+0.000	**0.664**	−0.015	**0.664**	+0.000
	AmbGCE	32	0.780	+0.000	0.927	−0.009	0.763	−0.034
	TanGCE	32	**0.696**	+0.000	**0.727**	−0.032	**0.664**	−0.002
Gemma-2-27B	AmbGCE	12	0.812	+0.000	0.890	−0.003	0.809	−0.005
	TanGCE	12	**0.680**	+0.000	**0.655**	−0.057	**0.716**	−0.002
	AmbGCE	23	0.764	+0.000	0.819	−0.003	0.818	−0.098
	TanGCE	23	**0.665**	−0.001	**0.659**	−0.028	**0.685**	−0.011
	AmbGCE	34	0.701	+0.000	0.889	−0.000	0.821	−0.014
	TanGCE	34	**0.670**	+0.000	**0.695**	−0.079	**0.725**	−0.006
Llama-3.2-1B	AmbGCE	4	0.790	+0.000	0.856	−0.001	0.790	−0.028
	TanGCE	4	**0.691**	+0.000	**0.565**	−0.030	**0.612**	+0.008
	AmbGCE	8	0.887	+0.000	0.901	+0.003	0.839	−0.039
	TanGCE	8	**0.704**	+0.000	**0.550**	−0.019	**0.660**	−0.017
	AmbGCE	12	0.815	+0.000	0.935	−0.001	0.853	−0.027
	TanGCE	12	**0.734**	+0.000	**0.755**	−0.012	**0.715**	−0.029
Llama-3.2-3B	AmbGCE	7	0.813	+0.000	0.903	−0.002	0.820	−0.016
	TanGCE	7	**0.728**	−0.001	**0.571**	−0.017	**0.649**	−0.007
	AmbGCE	14	0.909	+0.000	0.929	−0.001	0.860	−0.046
	TanGCE	14	**0.768**	+0.000	**0.610**	−0.010	**0.745**	−0.023
	AmbGCE	21	0.924	+0.000	0.945	−0.005	0.872	−0.027
	TanGCE	21	**0.747**	+0.000	**0.758**	−0.009	**0.733**	−0.030
Mistral-7B	AmbGCE	8	0.820	+0.000	0.895	−0.004	0.852	−0.020
	TanGCE	8	**0.730**	+0.000	**0.599**	−0.012	**0.692**	−0.005
	AmbGCE	16	0.935	+0.000	0.914	−0.003	0.884	−0.032
	TanGCE	16	**0.724**	+0.000	**0.596**	−0.028	**0.730**	−0.038
	AmbGCE	24	0.938	+0.000	0.926	−0.002	0.879	−0.039
	TanGCE	24	**0.721**	−0.001	**0.708**	−0.009	**0.748**	−0.018

Table 10: Supplementary deeper-layer sycophancy comparison at $\lambda = 1.0$ after 7 rounds, across layers 8, 16, and 23. Columns T (tangent-constrained) and A (ambient baseline) compare erasure methods. Utility is answer-matching change relative to clean representations. Leakage and displacement decrease from left to right (lower is better); utility increases from left to right (higher is better). This table is included as supplementary evidence of directional consistency with the layer-8 main-text comparison in Table 2. Deeper-layer tangent runs continue to reduce leakage more than ambient baselines across this sweep.

Layer	T leak	A leak	T util	A util	T disp	A disp
8	0.063	0.213	-20.7pp	-0.5pp	2.88	0.62
16	0.153	0.400	+0.0pp	+0.0pp	3.79	0.27
23	0.138	0.361	-0.1pp	-0.2pp	44.50	3.11

Table 11: Supplementary deeper-layer safety comparison at $\lambda = 1.0$ after 7 rounds, across layers 8, 16, and 23. Columns T (tangent-constrained) and A (ambient baseline) compare erasure methods. Utility is pairwise helpfulness change. Leakage and displacement decrease from left to right (lower is better); utility increases from left to right (higher is better). This table is included as supplementary evidence of directional consistency with the layer-8 main-text comparison in Table 2. Deeper-layer tangent runs continue to reduce leakage more than ambient baselines across this sweep.

Layer	T leak	A leak	T util	A util	T disp	A disp
8	0.0725	0.2583	-0.052	-0.023	2.70	0.18
16	0.1292	0.3517	-0.092	+0.003	4.37	0.60
23	0.1483	0.3404	-0.062	-0.152	38.32	5.30

Table 12: Local Table-2-style CelebA visual comparison using CLIP ViT-B/32 embeddings. For each target concept we intervene once per method and evaluate (i) concept leakage via retrained linear/nonlinear probes (lower is better) and (ii) mean control-task accuracy change across 3 least-correlated control attributes (closer to zero is better). Targets and controls: Male $\rightarrow$ Blurry,Narrow_Eyes,Straight_Hair; Smiling $\rightarrow$ Straight_Hair,Bushy_Eyebrows,Gray_Hair; Blond_Hair $\rightarrow$ Straight_Hair,Narrow_Eyes,Blurry. Lambdas follow Table 2: all methods λ=0.5 except TanGCE λ=8.0.

	Male			Smiling			Blond_Hair		
Method	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	.993	.994	-0.0pp	.891	.911	+0.0pp	.909	.933	-0.1pp
Random Per-Samp.	.993	.994	+0.0pp	.891	.907	+0.2pp	.907	.932	-0.1pp
Linear Proj.	.993	.992	+0.0pp	.900	.906	-0.3pp	.919	.931	-0.2pp
INLP	.994	.993	-0.0pp	.900	.903	+0.3pp	.918	.932	-0.0pp
LEACE	.994	.991	-0.0pp	.894	.894	+0.2pp	.909	.925	-0.1pp
AmbGCE	.990	.995	-0.2pp	.858	.884	-0.4pp	.893	.920	+0.2pp
TanGCE	.887	.953	-0.8pp	.649	.649	-0.6pp	.699	.747	-2.1pp

Table 13: Per-concept nonlinear probe leakage on CelebA (CLIP ViT-B/32), all 40 binary attributes. Each cell is retrained nonlinear probe test accuracy after erasure (lower = stronger erasure). λ^* is selected per method per concept from $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation NL accuracy. Bold marks the lowest leakage per row. Abbrev.: RandG = Random Global, RandS = Random Per-Sample, LinProj = Linear Projection, AmbGCE = ambient gradient-based concept erasure, TanGCE = tangent gradient-based concept erasure.

Concept	RandG	RandS	LinProj	INLP	LEACE	IGBP	AmbGCE	TanGCE
5_o_Clock_Shadow	.92	.92	.91	.92	.87	.92	.91	**.77**
Arched_Eyebrows	.79	.79	.76	.76	.69	.77	.77	**.59**
Attractive	.73	.74	.69	.73	.63	**.53**	.73	.60
Bags_Under_Eyes	.76	.75	.74	.75	.70	.77	.75	**.69**
Bald	1.00	.99	.99	.99	.98	1.00	.99	**.97**
Bangs	.91	.91	.89	.89	.87	.92	.91	**.68**
Big_Lips	.80	.79	.74	.80	**.68**	.82	.81	.73
Big_Nose	.76	.76	.73	.74	.72	.73	.76	**.61**
Black_Hair	.85	.84	.82	.81	.79	.83	.82	**.46**
Blond_Hair	.92	.92	.94	.92	.92	.92	.92	**.72**
Blurry	.98	.98	.98	.96	.97	.98	.97	**.88**
Brown_Hair	.80	.80	.76	.75	.75	.80	.80	**.52**
Bushy_Eyebrows	.87	.88	.86	.85	.76	.87	.86	**.66**
Chubby	.94	.94	.95	.95	.92	.95	.94	**.81**
Double_Chin	.94	.94	.94	.93	.92	.94	.95	**.85**
Eyeglasses	.99	1.00	.96	.93	.96	1.00	.96	**.90**
Goatee	.96	.97	.97	.93	.96	.96	.95	**.85**
Gray_Hair	.97	.97	.97	.97	.97	.97	.98	**.89**
Heavy_Makeup	.89	.89	.89	.86	.85	.88	.88	**.70**
High_Cheekbones	.79	.79	.77	.75	.70	.57	.76	**.50**
Male	.99	.99	.99	.99	1.00	.98	1.00	**.85**
Mouth_Slightly_Open	.89	.89	.89	.71	.72	.66	.80	**.43**
Mustache	.94	.95	.95	.94	.92	.95	.93	**.77**
Narrow_Eyes	.88	.88	.80	.87	.80	.88	.88	**.78**
No_Beard	.94	.95	.94	.93	.92	.94	.95	**.77**
Oval_Face	.70	.71	.69	.68	**.67**	.69	.67	.72
Pale_Skin	.93	.93	.94	.95	.93	.92	.93	**.83**
Pointy_Nose	.75	.72	.71	.69	**.63**	.72	.75	.64
Receding_Hairline	.94	.94	.96	.96	.94	.94	.95	**.69**
Rosy_Cheeks	.93	.93	.93	.94	.91	.94	.93	**.79**
Sideburns	.95	.95	.94	.94	.95	.94	.94	**.82**
Smiling	.90	.89	.91	.74	.82	.59	.88	**.44**
Straight_Hair	.78	.78	.79	.77	.78	.78	.78	**.49**
Wavy_Hair	.78	.76	.77	.77	.78	.77	.78	**.52**
Wearing_Earrings	.90	.89	.86	.88	.85	.89	.88	**.61**
Wearing_Hat	1.00	1.00	.97	.97	.97	1.00	.97	**.94**
Wearing_Lipstick	.90	.90	.90	.89	.85	.88	.92	**.75**
Wearing_Necklace	.84	.85	.80	.80	.77	.85	.83	**.68**
Wearing_Necktie	.93	.93	.92	.92	.91	.92	.93	**.78**
Young	.86	.86	.85	.84	.83	.86	.87	**.60**
Mean	.88	.88	.87	.86	.84	.86	.87	**.71**

Table 14: Per-concept mean control accuracy change ΔCtrl (%) on CelebA, all 40 binary attributes. For each concept, ΔCtrl is averaged over the 5 least-correlated control attributes (closer to zero = more surgical). λ^* selection matches Table 13.

Concept	RandG	RandS	LinProj	INLP	LEACE	IGBP	AmbGCE	TanGCE
5_o_Clock_Shadow	+0.0	+0.1	-0.7	+0.1	+0.0	+0.1	-2.3	-0.2
Arched_Eyebrows	-1.3	-2.3	-1.2	-0.8	-1.1	-2.5	+0.6	-0.6
Attractive	-0.3	-1.0	-1.7	-0.8	-0.3	-0.9	+0.0	-0.8
Bags_Under_Eyes	-3.7	-1.7	-0.3	-7.7	-0.3	-0.3	-0.3	-0.1
Bald	+0.1	+0.0	-0.1	+0.1	-0.1	+1.1	+2.3	+0.2
Bangs	-3.7	-1.2	-0.5	-7.9	-0.1	-0.7	-0.3	-0.8
Big_Lips	+0.1	-0.0	-0.3	+0.0	-0.1	-0.5	+0.1	-0.1
Big_Nose	-0.1	+0.3	-3.5	-9.7	+0.0	-0.3	-1.8	+0.1
Black_Hair	-0.1	-0.0	+0.1	+0.2	+0.3	-0.3	-0.3	-0.7
Blond_Hair	-0.3	+0.1	-1.4	+0.7	-0.1	-0.4	-1.1	-1.6
Blurry	-1.3	+0.0	-0.1	-5.5	+0.1	-0.7	+0.7	-0.1
Brown_Hair	+0.1	-0.9	-0.3	-6.0	-0.7	+0.4	+0.1	-0.2
Bushy_Eyebrows	+0.0	+0.1	-0.1	-1.5	+0.2	-0.9	-0.3	-0.5
Chubby	-0.1	+0.3	-0.2	-1.4	-0.2	-0.2	+0.4	-2.0
Double_Chin	+0.1	+0.1	-0.4	-12.7	+0.3	+0.7	-0.7	-2.5
Eyeglasses	-0.1	+0.1	-0.0	-0.1	-0.2	-0.4	-0.2	+0.2
Goatee	+0.1	-0.1	+0.0	+0.2	+0.1	+0.5	-0.3	-0.4
Gray_Hair	-0.2	-0.3	-2.1	+0.4	-0.1	-1.1	-1.5	-1.5
Heavy_Makeup	-0.3	-0.1	-1.1	-1.7	-1.3	-1.3	-1.1	-0.9
High_Cheekbones	+0.1	+0.2	+0.1	-0.3	-0.1	-0.4	-0.0	-0.7
Male	-0.3	-0.1	-0.7	-0.2	-0.3	-2.2	-0.9	-1.1
Mouth_Slightly_Open	-0.3	-0.6	+0.1	+0.4	+0.2	+0.3	+0.1	-0.2
Mustache	+0.0	-0.1	+0.1	-10.3	-0.5	+1.0	+0.5	-1.5
Narrow_Eyes	-0.1	-0.2	-0.3	+0.2	-0.1	+0.3	+0.0	+0.0
No_Beard	-1.2	-0.2	-4.0	-6.5	-1.6	-0.1	-0.7	-0.5
Oval_Face	-2.0	+0.1	-0.1	-14.4	-0.6	-0.3	+0.2	-0.3
Pale_Skin	-1.2	-0.1	-0.1	-6.8	-0.9	-0.1	+0.7	-1.0
Pointy_Nose	-0.1	-0.2	+0.3	-1.1	+0.2	-0.7	-0.1	-0.1
Receding_Hairline	-0.1	-0.1	-0.1	-5.9	+0.0	-0.3	+0.1	-1.1
Rosy_Cheeks	-0.5	+0.1	-0.6	-0.1	-0.2	-0.1	-0.1	-1.3
Sideburns	-0.4	+0.0	-0.1	+0.1	+0.1	-0.6	-2.5	-0.1
Smiling	-0.3	-1.3	+0.1	+0.6	+0.4	+0.1	+0.1	-0.1
Straight_Hair	-0.1	-0.2	-0.1	-0.9	-0.1	-0.8	+0.5	-0.7
Wavy_Hair	+0.1	-0.7	-0.9	+0.2	+0.1	-0.4	-0.1	-0.7
Wearing_Earrings	+0.8	-0.3	-1.6	-0.3	-0.1	-1.3	-0.5	-0.3
Wearing_Hat	-0.1	+0.0	-0.1	-0.6	+0.0	+1.0	+0.1	-0.2
Wearing_Lipstick	+0.2	-0.1	-6.1	-21.5	-1.9	-1.2	-1.4	-0.9
Wearing_Necklace	+0.1	-1.7	-1.7	-9.3	+0.3	-1.8	+0.2	-1.2
Wearing_Necktie	+0.0	+0.1	+0.0	+0.2	-0.2	-0.7	-13.4	-1.3
Young	-0.0	-0.7	-1.7	-10.7	+0.1	-0.9	-2.2	-0.8
Mean	-0.4	-0.3	-0.8	-3.5	-0.2	-0.4	-0.6	-0.7

Table 15: Geometry evidence map organizing the appendix diagnostics into four research blocks. Each block addresses a distinct aspect of manifold structure: subspace sufficiency examines whether task-relevant computation concentrates in a low-dimensional linear subspace, perturbation sensitivity tests whether on-manifold directions matter more than orthogonal directions, representation connectivity evaluates whether the manifold remains connected across domains, and multi-layer accumulation tracks error compounding through the network. These diagnostics support the mechanism discussion and motivate the need for manifold-aware erasure methods but do not by themselves establish causality.

Experiment Block	Research Question	Primary Evidence
Subspace sufficiency	Subspace concentration	Top-1 agreement under intrinsic projection
Perturbation sensitivity	High-variance vs. orthogonal directions	KL and top-1 under on/off-subspace noise
Representation connectivity	Shared vs. domain manifolds	Cross-domain interpolation smoothness
Multi-layer accumulation	Error compounding across the network	Simultaneous intrinsic projection across layers